LESSON 3.8
SOLVING MULTI-STEP, REAL-WORLD PROBLEMS WITH FRACTIONS OR DECIMALS – PART 1

LESSON INTRODUCTION

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

LEARNING TARGETS

- I can solve three-step, real-world problems with decimals or fractions.

BENCHMARK COHERENCE

BUILDING ON

MA.5.AR.1 Solve problems involving the four operations with whole numbers and fractions.

WORKING TOWARD

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

ESSENTIAL QUESTIONS

- What are the similarities and differences between solving two-step and three-step real-world problems?
- How can you tell what steps and operations are needed to solve three-step problems?

LESSON NARRATIVE

In this lesson, students continue to solve real-world, multi-step word problems involving rational numbers, using the problem-solving skills they developed in earlier lessons from this unit. After a Warm-Up Career Profile Video, students are guided through problem solving with three-step real-world problems involving decimals and fractions.

COMPONENT SUMMARY

3.8.1 WARM-UP *(~5 MIN.)*

Career Profile Video – Structural Engineer

3.8.2 GUIDED INSTRUCTION *(15-20 MIN.)*

Guided instruction on problem solving with three-steps

3.8.3 YOUR TURN! *(15-20 MIN.)*

Practice solving three-step real-world problems with rational numbers

3.8.4 WRAP-UP *(~5-10 MIN.)*

Students rate whether they agree, disagree, or are unsure about statements and justify their thinking

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- Career Profile Video
- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- 3.8.1 Warm-Up contains a 3-minute video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may create inaccurate numeric expressions.
- Students may make computational errors when performing operations on multi-digit decimal and fraction problems.
- Students may have difficulty identifying the steps needed to solve problems, and in which order to solve the steps to find the correct solution.

THINGS TO CONSIDER

- To foster flexible thinking and problem-solving strategies, ask students to draw a visual model first and then create a numeric expression from the visual model.
- Provide students who may need assistance organizing information a graphic organizer and/or annotation strategies while engaging with three-step problems.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

Utilize Think-Pair-Share opportunities in 3.8.2 Guided Instruction and 3.8.3 Your Turn! for students to practice articulating their strategies and approaches in problem-solving in order to continue building fluency in real-world contexts.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Throughout the lesson, students will be reading, analyzing, and discussing real-world scenarios to determine the appropriate mathematical operation to apply for solving problems. Students use concrete and abstract models to investigate real-world problems and find solutions that are reasonable given the context.

DIFFERENTIATE INSTRUCTION

Consider using a jigsaw for 3.8.3 Your Turn! practice problems. Students can be assigned to work on one of the five problems, then rotate to new groups where one student is responsible for sharing thinking and answers for the problem as the other students work on it.

Reading comprehension is at the forefront of learning for real-world problems.

Consider your students when planning for 3.8.4 Wrap-Up – this might work for your students if they need independent processing time, or this activity could be transformed into a Socratic seminar-style debate with evidence required in collaboration with an English teacher.

3.8.1 WARM-UP: CAREER – BODY STRUCTURES ENGINEER

Component Overview: *Display the "Body Structures Engineer" video for the entire class. After watching this 3-minute video, prompt the students to read and answer the questions in the Warm-Up independently or in partners.*

QUESTIONING

Do you know someone who is a structural engineer or works in engineering? Consider bringing in a visitor for your class to ask questions and learn from. Additionally, consider having students return to this lesson later in the year for project-based learning around this Career Profile.

3.8.1 Warm-Up: Career – Body Structures Engineer

Rebecca Lees, a body structures engineer from Jaguar Land Rover, works on high-profile projects and talks about how she came to pursue a career in automotive, her educational background, and how an international work placement played a key role in setting her on a path to her career.

1. **What did you find interesting about body structures engineers?**
   Answers will vary. Rebecca Lees talks about getting to see all the proto-types. She discusses getting to test the products while they're in camouflage.

2. **Structural engineers focus on the structure of a building or a bridge. The structures should withstand the stresses and pressures of the environment. What stresses or pressures should the body of a car be able to withstand?**
   Answers will vary. Wind, rain, snow, sun, heat, cold

3.8.2 Guided Instruction: Three-Step Rational Problems

1. For his recital, a student practiced the piano 2.18 hours each day for 5 days last week. This week, he practiced 0.75 hours each day for 4 days. How many hours did he spend practicing? 13.9 *hours*

2. The Across Florida 200 Trail Run is an event where participants run 200 miles of trails between the Withlacoochee Bay Trail and Crescent Beach. To prepare for the Across Florida 200 Trail Run, Walt ran 2.18 hours each day for 5 days last week. This week, he ran 2.75 hours each day for 4 days. Walt runs at an average pace of 6.23 miles per hour.
   a. How many hours has Walt run over the last two weeks? 21.9 *hours*
   b. How many miles has Walt run over the last two weeks? 136.437 *miles*
   c. Based on Walt's average pace, how much longer would it take him to run the Across Florida 200 Trail Run than the amount of time he has spent practicing the last two weeks? ≈ 10.2 *hours longer*

3. The Across Florida 200 Trail Run offers a team competition where a team of 4 runners take turns to run the 200 miles.
   A team of runners and their average pace is shown.

Name	Molly	Reginald	Tyler	Liz
Average Pace (miles per hour)	6.75	6.75	8.2	9.58

To prepare for the race the team ran part of the trail.

Team Member	Description	Miles
Tyler	Withlacoochee Bay Trail to Lake Rousseau Dam & Recreational Area	10
Liz	Lake Rousseau to Florida Trail	15
Molly	Florida Trail Part 1	35
Reginald	Florida Trail Part 2	25

How long did it take the team to run the practice run of 85 miles? 11.7 *hours*

3.8.2 GUIDED INSTRUCTION: THREE-STEP RATIONAL PROBLEMS

Component Overview: *Students practice planning and solving three-step rational problems. Encourage students to recall strategies used to solve one- and two-step real world problems in previous lessons to build upon background knowledge. Whole group and or small group work can be incorporated in this activity to provide differentiation for different types of learners.*

TEACHER MOVES

Think-Pair-Share

Providing students with ample processing time using Think-Pair-Share strategy gives both independent and verbal processing for more precise and confident understanding.

DIFFERENTIATE INSTRUCTION

Real-world problems require reading and comprehension. Consider pairing with a reading teacher for interdisciplinary support. Students can get additional support in identifying key features of the text that will lead to better comprehension, in addition to more practice time. For example, if students use a specific reading comprehension strategy with another teacher, consider having students apply that strategy in these contexts.

ENRICHMENT

Consider prompting questions to guide students toward key features of the text.

- What is the question asking? (total time ran by all)
- How will we know how long they took? (add up how long each person took)
- How do we know how long each person will take? (multiply their pace times total miles run)

3.8.3 YOUR TURN!

Component Overview: *The unit has used scaffolding to help students become successful at solving problems using rational numbers. Fluency with rational values is an important focus for 6th grade. Teachers may consider allowing students to use a calculator to answer the questions.*

TEACHER MOVES

Notice that “half” is not in fraction form. Students might just add or multiply the first two terms they see without considering the “half” of blueberries that spoiled.

QUESTIONING

For students who finish early, challenge them to determine after how many days the pool would be empty if the leak was not fixed (58 days), and to write an expression that could represent that value.

ENRICHMENT

Did you subtract each type of wing from 2? Or did you add up all types before subtracting that value from 2? Would you do this differently next time? Why or why not?

DIFFERENTIATE INSTRUCTION

Differentiate the content to engage more learners by giving students the opportunity to substitute question 5 with their own question and self-made context. They must write the question as well as verify the values chosen by finding the solution.

3.8.3 Your Turn!

1. On Sunday, a farmer picked $1\frac{3}{4}$ barrels of blueberries. On Monday, the farmer picked $\frac{1}{5}$ as many blueberries as they did on Sunday. On Tuesday, the farmer discovered that half of his barrels of blueberries were spoiled. How many total barrels of blueberries did the farmer have left to sell? $1\frac{1}{20}$ *barrels*
2. A pool that holds 273.68 gallons of water lost 4.79 gallons of water each day while there was a leak that went unnoticed. Tad noticed the water was lower, so he added 20.45 gallons of water before he realized there was a leak on the side. If Tad fixed the leak after 8 days of leaking, how many gallons of water does he currently have in the pool? 255.81 *gallons*
3. A paving company ordered $4\frac{3}{4}$ tons of cement to pave the streets of a new neighborhood. Additionally, the company ordered $7\frac{2}{3}$ tons of cement to pave the guest parking lot for the pool area at the front of the neighborhood. After pouring, they only needed $\frac{2}{3}$ of the cement they ordered. How many tons of cement were left unused? $4\frac{5}{36}$ *tons*
4. A grocery store opens a new wing bar that charges by the pound. Yasmine wants to purchase $\frac{1}{4}$ pound of garlic parmesan wings, $\frac{1}{8}$ pound of mild wings, and $\frac{2}{5}$ pound of teriyaki wings. She also wants to purchase some hot wings. If Yasmine wants to purchase a total of 2 pounds of wings, how many pounds of hot wings should she buy? $1\frac{9}{40}$ *pounds*
5. Rawslyn is creating study kits for her students. Each folder costs $1.79 and each copy of the study guide costs $0.95. Each student gets one of each and there are 22 students in the class. If Rawslyn has $35.43 in her wallet, how much more money will Rawslyn need to create the study kits? $24.85

3.8.4 Wrap-Up: Cool Down

For each statement, write whether you agree with it, disagree with it, or are unsure.

Statement	Agree/Disagree/Unsure
The more digits a number has, the larger it is.	*Answers will vary. Unsure, it depends on if the number is a whole number or a decimal number.*
When you add two positive decimals together, the value is always larger.	*Answers will vary. Agree, the decimal number will grow.*
To add fractions, their denominators must be the same.	*Answers will vary. Agree, to add fractions, they must first be like fractions.*
The product of two fractions is always a larger number.	*Answers will vary. Disagree, when you multiply fractions the product is smaller than the factors.*
When you put a zero on the end of any number, it makes it larger.	*Answers will vary. Unsure, it depends on whether the number is a whole number or a decimal number.*

3.8.4 WRAP-UP: COOL DOWN

Component Overview: *This Wrap-Up assesses students' place value and arithmetic reasoning in relation to rational numbers and operations. Students record whether they agree, disagree, or are unsure about the statements.*

DIFFERENTIATE INSTRUCTION

Differentiating by Product:

- Link Up: Students write whether they agree or disagree on a whiteboard or use their hand to indicate. Students then must "Link Up" and find someone who does not have the same answer and try to convince them why their answer is correct.
- A/S/N: Students must include "Always," "Sometimes," or "Never" in their responses. The first and last prompts will be the "Sometimes" in which students must engage in mathematical discourse.
- Prove It: Challenge students to provide evidence for each answer with a justification. Correct answers will only be accepted if there is an example. Counterexamples included will be worth bonus points!